Docs

Description

This is a project using uMyo EMG sensors to make a wristband, which sends data to an iPhone via Bluetooth, which is then used to perform gesture classification.

Each gesture can then be paired with an action. An action can be any API call that can be performed by the phone, such as HomeKit and Shortcuts APIs.

Milestones

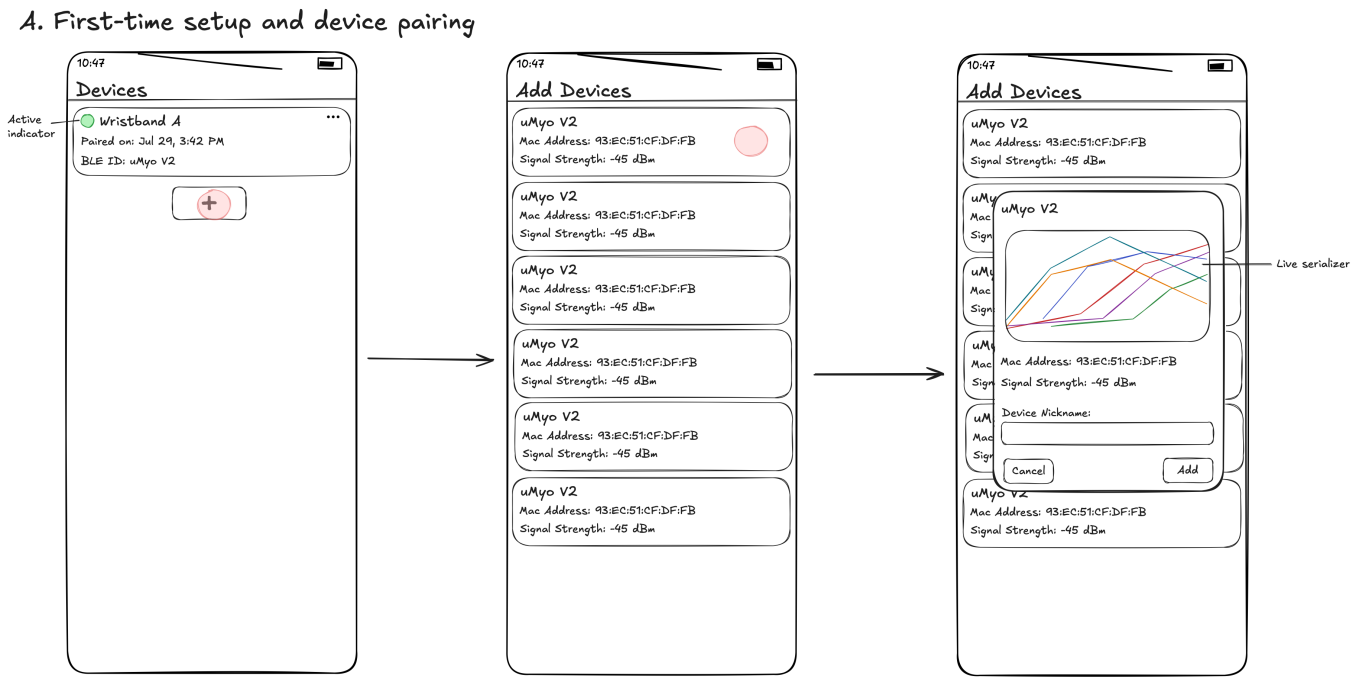
V1: Limit to basic gesture recognition flow, no API calls yet. Only allow one model at a time -- new model rewrites old model.

V2: Build out action assignment flow.

V1 Core User Flows:

A. **First-time setup and device pairing**:

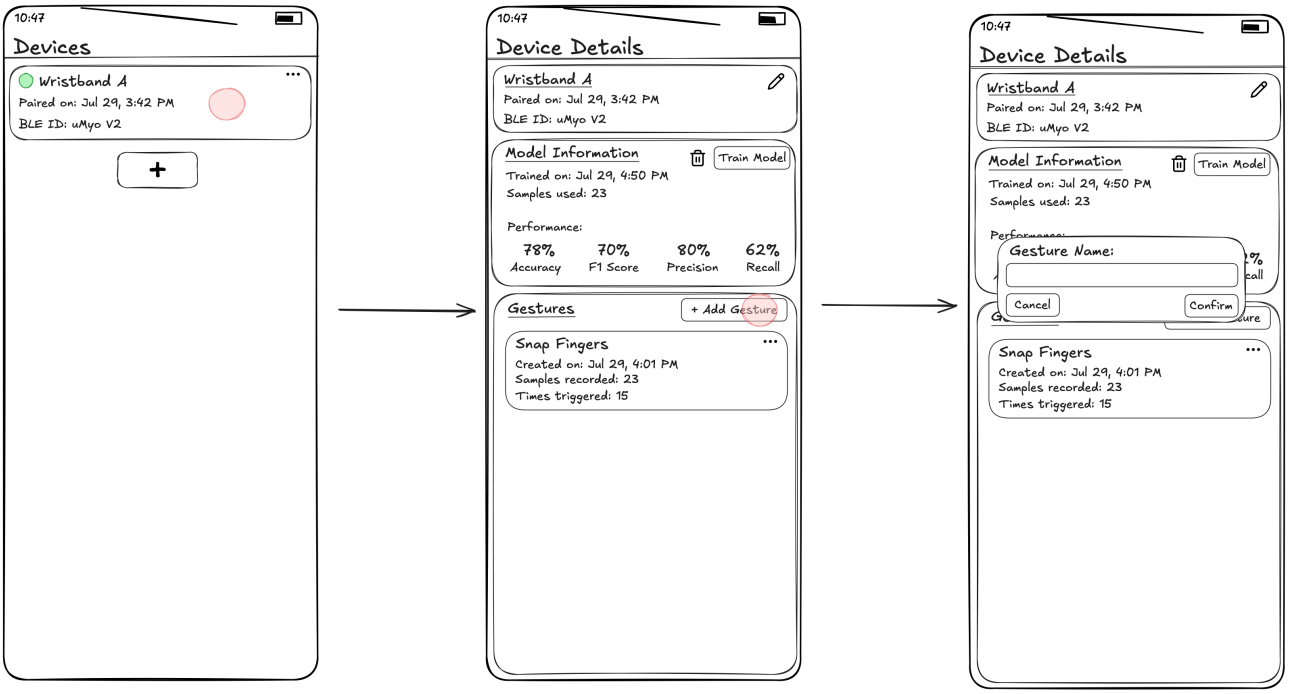
1. Home page shows a list of paired devices with a button to add a device.
    - Maybe show which device is active too right now, limit to one active device at a time.
2. Clicking on the button navigates to "Add Devices" page for BLE device discovery.
    - Maybe good to limit results to uMyo devices only.
3. Clicking on a discovered device opens a pop-up with buttons to cancel or add the device along with a field to input a device name.
    - Optional: provide a serializer in the same pop-up to show that the device is transmitting data.



B. **Gesture creation**:

1. Gestures are specific to a device. After a device is paired, user can click on the device to show details and create a gesture.
2. Clicking on "Create Gesture" opens a pop-up requesting a name (e.g., "snap fingers") and a button to confirm.

B. Gesture creation

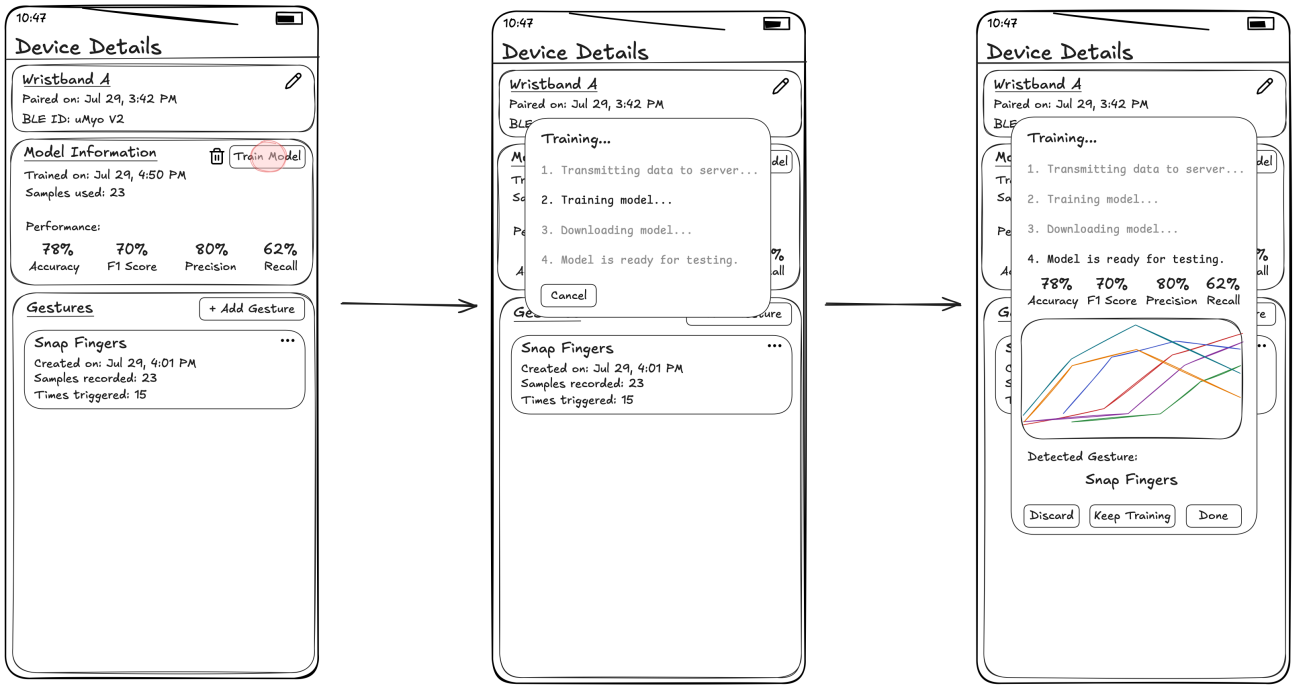


- **Gesture sample recording**:
  1. Once gesture is created, we navigate to the gesture details page which includes a section to show training samples along with a button to start recording samples.
  2. Clicking "Record Sample" shows a pop-up with a 3 seconds countdown before it begins recording. Once it begins, it will show a live serializer and record for 3 seconds.
  3. Once a sample is recorded, the live visualizer is replaced with a static waveform of the recording and three buttons: "Discard", "Retry", or "Keep".
  4. If user selects "Keep", the sample is saved and the pop-up is closed.
     - If a user wants to record multiple samples in a row, they can just click Record Sample again. This might be jarring, but we can revisit later.

C. Gesture sample recording

- **Model training**:
  1. Once there is at least 10 training samples, user can go back to the device details page and click on "Train Model".
     - We are training a single multi-class model so it can distinguish between all gestures + a "none".
  2. The app will call an external endpoint and export the training data to train the model.
  3. For simplicity, our first version will just make the user wait until training is done.
  4. Once training is complete, user will be shown model performance metrics and three buttons: "Discard", "Keep Training", or "Done" along with a live serializer and a "Detected Gesture" for live testing.
  5. Clicking on "Done" will save the model on-device and allow the user to activate a device for real-time gesture recognition.

D. Model training

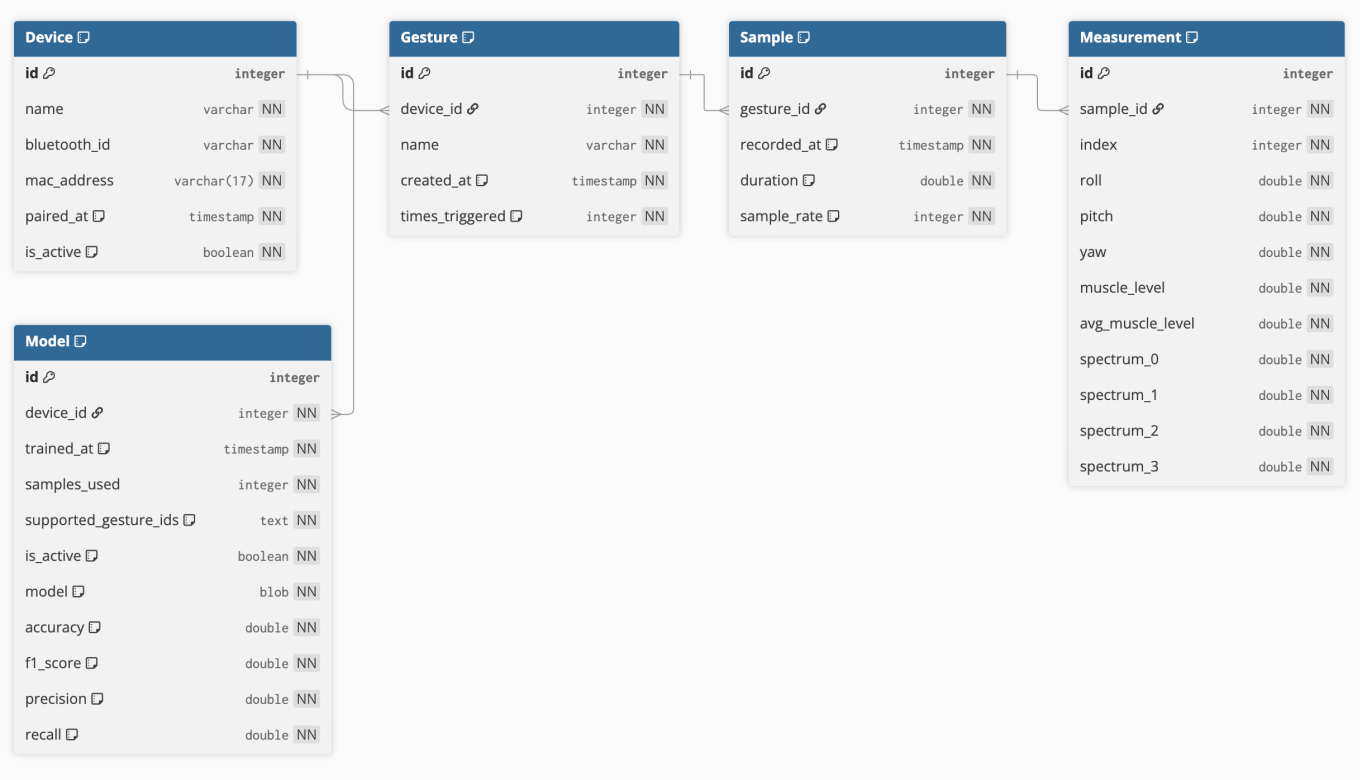


Discussions:

- **On-device training?**: So Apple has two ML frameworks: CreateML and CoreML. CreateML is Apple's AutoML solution (automates things like feature engineering, algorithm selection, and hyperparameter tuning) to training models, while CoreML is the inference framework for running models on-device. CreateML provides several model types out of the box, but not all of them can be trained on mobile devices (iOS, iPadOS, visionOS). The model we would need for this type of application (time-series data) would be the MLActivityClassifier which can only be trained on macOS. So, no on-device training is possible.

- **CreateML vs Sklearn**: At the end of the day, we need a CoreML model to run on-device. We can achieve this by creating the model using CreateML or convert an Sklearn model using Apple's library "coremltools". Since we have to train this model off-device anyway, Sklearn would be more flexible since it can be trained on any device, not just Macs. It's probably also easier to achieve better performance since we have more control over the model (and I'm much more familiar with it). Therefore, we'll be training models using Sklearn.

Data Model



Device

Represents a paired uMyo device.

Relationships:

- Has many gestures
- Has one model

Properties:

- id | pk
- name
- bluetooth_id
- mac_address
- paired_at
- is_active

Gesture

Represents a user-defined gesture, specific to a device.

Relationships:

- Belongs to one device
- Has many training samples

Properties:

- id | pk
- device_id | fk
- name
- created_at
- times_triggered

Sample

Represents a single training sample for a gesture. One training sample contains many measurements (depending on sample rate).

Relationships:

- Belongs to one gesture
- Has many data points

Properties:

- id | pk
- gesture_id | fk
- recorded_at
- duration
- sample_rate

Measurement

Represents a single raw measurement (data point) for a training sample.

Relationships:

- Belongs to one training sample

Properties:

- id | pk
- sample_id | fk
- index
- roll
- pitch
- yaw
- muscle_level
- avg_muscle_level
- spectrum_0
- spectrum_1
- spectrum_2
- spectrum_3

Model

Represents a trained multi-class classification model using samples from all gestures for a specific device. In v1, we won't be able to view/switch to old models, but eventually one device can have multiple models

with one active at a time.

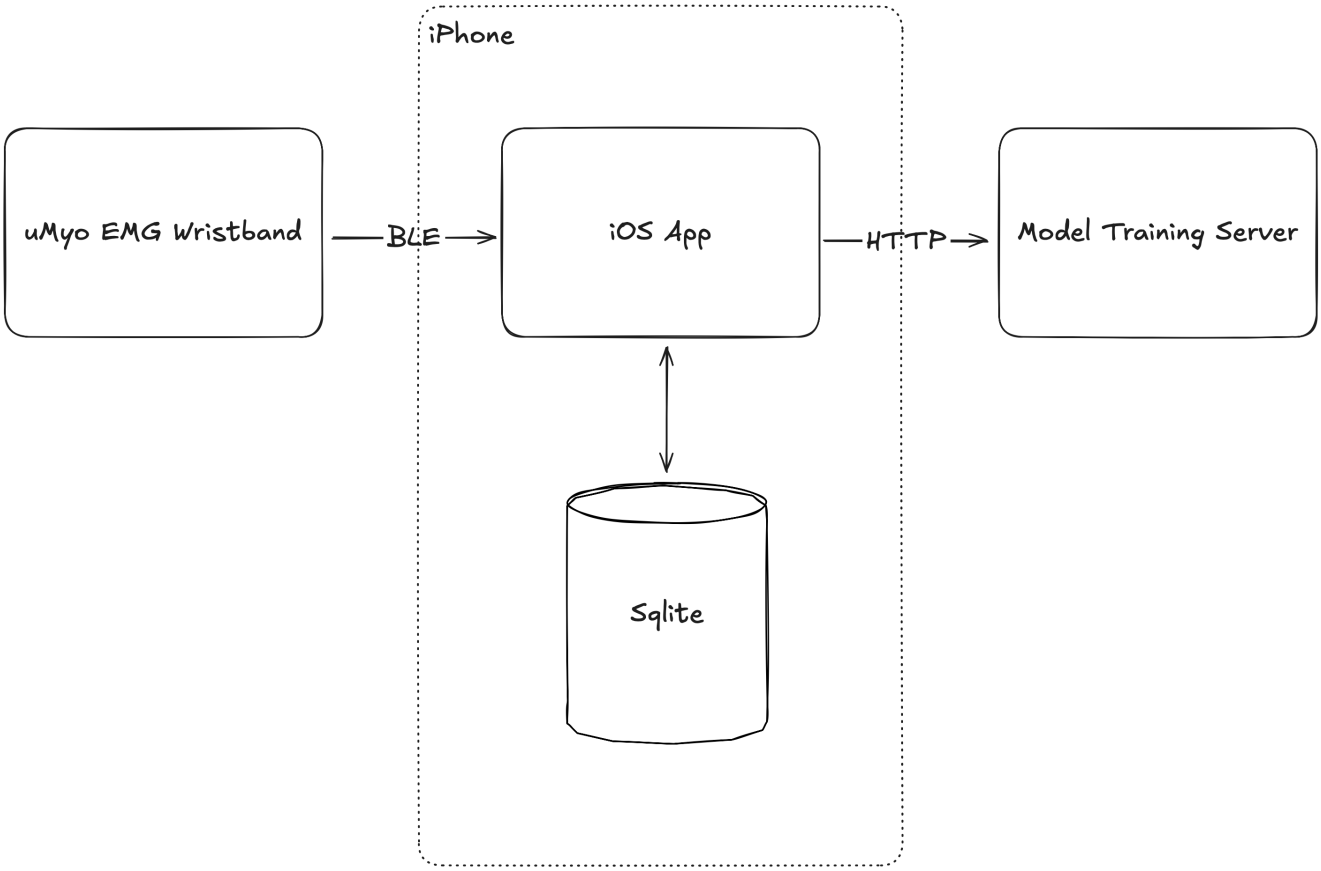
Relationships:

- Belongs to one device

Properties:

- id | pk
- device_id | fk
- trained_at
- samples_used
- supported_gesture_ids // not a fk or junction table because we just need to know which gestures this model supports
- is_active
- model
- accuracy
- f1_score
- precision
- recall

Technical Architecture



The architecture for this project is fairly simple. All of the functionalities except for model training runs on-device, so we just need a client and a server with a training endpoint.

Endpoints

```
POST /train-model
- Body: training data
- Returns: CoreML model + metrics

GET /health
- Health check
```

uMyo Advertisement Data Packet Layout

Notes:

- The first x bytes are skipped until we find the first 0x08 (byte value of 8). Byte 0 below is byte x + 1.
- Comments that indicate how something is used is based on uMyo's Arduino Library for parsing uMyo's BLE data on nRF52x and ESP32-based Arduinos.
- Discussion around spectrum calculation is based on uMyo's firmware.

Byte	Content	Comment	Parsing Notes
0	8 (0x08)	Start of header/useful data. Not meant to be used as the ASCII character '8', this is the decimal 8.	
1	'u' (0x75)	These next ones are ASCII characters. Perhaps useful to filter out useless packets?	
2	'M' (0x4D)		
3	'y' (0x79)		
4	'o' (0x6F)		
5+	255 (0xFF)	Location seems to vary — there maybe a gap between 'o' and 0xFF. while(pack[pp] != 255 && pp < len) pp++;	
Next	adc_id	This is the packet sequence id — telling us the correct order of packets.	
Next+1	batt_level	Battery level. To get the battery mv do: 2000 + batt_level*10	
Next+2	sp0	Spectrum[0]. This is sent as 8-bits but it's actually sent at reduced precision and must be left shifted to 16-bits. Along with sp1, not really used to calculate anything. The uMyo device performs FFT that takes in 8 input samples and produces 4 frequency bins (spectrum). It seems that the later bins (2 and 3) represent higher frequencies than the earlier bins (0 and 1) — search for high_sp in main.c.	Left-shift 8 bits. int16_t sp0 = pack[pp++]<<8;

Byte	Content	Comment	Parsing Notes
Next+3	muscle_avg	Device-calculated average of muscle activity level. Shouldn't be used directly; these are used to calculate device_avg_muscle_level. Not the same as the user-calculated getMuscleLevel using spectrums.	
Next+4-5	sp1	Spectrum[1]. Along with sp0, not used to calculate anything.	2 bytes in big-endian format. Left-shift 8 bits. int16_t sp1 = (pack[pp]<<8)
Next+6-7	sp2	Spectrum[2]. Used with sp3 to calculate muscleLevel.	2 bytes in big-endian format. Left-shift 8 bits.
Next+8-9	sp3	Spectrum[3]. Used with sp2 to calculate muscleLevel. Valued twice as much as sp2. float lvl = devices[devidx].cur_spectrum[2] + 2*devices[devidx].cur_spectrum[3];	2 bytes in big-endian format. Left-shift 8 bits.
Next+10-11	qw	W component of the quaternion. A quaternion is a mathematically convenient alternative to euler angle representation for 3D rotations and is made up of four parts {x, y, z, w}. We can use these directly in our models or to compute roll, yaw, and pitch first.	2 bytes in big-endian format. Left-shift 8 bits.
Next+12	qx	X component of the quaternion.	Left-shift 8 bits.
Next+13	qy	Y component of the quaternion.	Left-shift 8 bits.
Next+14	qz	Z component of the quaternion.	Left-shift 8 bits.